**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 263 · LMC branching and debugging

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does STA do?

Q2. What does LMC HLT do?

Q3. Why must a loop's counter change?

Q4. Full adder: inputs A=1, B=0, carry-in=1. Find sum and carry-out.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

BRA — Branch Always

BRZ — Branch if Zero

BRP — Branch if non-negative

mailbox — LMC memory cell

### New learning

Understand — BRP tests whether ACC is non-negative; BRZ tests zero; BRA is unconditional.

Trace — Load the counter, output it and subtract one. Test BRP immediately: save the new value and loop only when it is non-negative.

Explain — After outputting 0, SUB ONE underflows; BRP is not taken, so HLT runs without storing a negative mailbox value.

### Worked example

Trace BRZ when input is 0.

1. Predict first — INP sets ACC to 0

2. Check next state — BRZ ZERO takes branch

3. Check next state — LDA ONE loads 1

4. Check next state — OUT displays 1.

# We Do · Practise with guidance

Trace the same code with input 4.

```
INP
BRZ ZERO
OUT
HLT
ZERO LDA ONE
OUT
HLT
ONE DAT 1
```

Predict first

INP makes ACC 4

Check next state

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BRZ not taken

OUT prints 4

HLT stops execution.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define BRA, BRZ and BRP and explain their effect on PC. [4]

### Question 2

Input 3: trace INP; BRZ STOP; SUB ONE; OUT; HLT; ONE DAT 1; STOP HLT. [6]

### Question 3

Explain why an unconditional backward branch without a tested exit can run forever, and provide a safe counter-based solution. [6]

## Exit Ticket · Answer before checking

An unconditional backward branch always terminates.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Copies ACC to mailbox.

Q2. Stops execution.

Q3. Without updating a tested state, the loop may not terminate.

Q4. $1+0+1=10_2$; sum=0, carry-out=1.

### You Do pages · Independent questions

Q1. BRA unconditional [1]; BRZ if ACC zero [1]; BRP if ACC non-negative [1]; taken branch changes instruction address [1].

Q2. ACC 3 [1]; BRZ not taken [1]; subtract 1 -> 2 [2]; OUT prints 2 [1]; HLT [1].

Q3. BRA repeats [1]; no progress/exit [1]; initialise count [1]; decrement [1]; store update [1]; test an exit condition [1].

### Exit Ticket · Model answer

False. It needs a reachable exit condition.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**